

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
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(1) Research Program/Project/Study Title:	Utilization of Nanochitosan for Bread Products
Proponent Name:	Dr. Ace D. Barredo
Academic Rank:	none
Co-Authors (with academic rank):	none
Campus/Satellite College/Address:	CapSU – Main Campus
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	Project 1: Utilization of Nanochitosan for Pandesal Project 2: Utilization of Nanochitosan for Ensaymada
Component Leader:	none
Academic Rank:	none

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation							
Municipality/City:	Roxas City						
Province:	Capiz						
Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
☐ Basic	☐ Pilot Testing						
☒ Applied	☐ Tech. Promotion/Commercialization						
(7) Sector/Commodity Agriculture/Fruit and Vegetables	(8) Discipline/Field of Specialization Food and Development						

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

All people frequently include bread goods in their meals or snacks, but Filipinos especially do so since we enjoy eating bread products like pandesal with coffee in the morning or ensaymada with juice for snacks. According to Antonia DeBianchi a lifestyle and food writer from South Florida, Feb 2, 2021 based on her article "How to store bread and never throw away a moldy loaf again", For all the joy bread brings to a table, it has a considerable weakness: A short shelf life. If you're not worrying about your favorite loaf growing mold, you're concerned about it getting stale. Regardless if it's store-bought, bakery-fresh or homemade, bread stays fresh longer when it's in a relatively air-tight environment since circulation speeds up the staling process, says Atlanta chef Jennifer Hill Booker (<https://www.insider.com/guides/kitchen/how-to-store-bread>).

Most of the time, bread goods are not stored properly, which causes bread to spoil easily or show signs of mold growth, requiring disposal. We need to look at the right and effective manner to store bread in order to maintain its freshness and shelf-life in order to address food waste issues involving bread. In this premise, the researcher would like to investigate the effect of nano chitosan powder on the shelf-life preservation of bread products. According to Sadik Büyükyörük, in his study entitled "Chitosan for Using Food Protection" As a compound obtained using various methods, the most prominent features of chitosan are attributable to its antimicrobial and antioxidant properties. Among all the antibacterial compounds from crustaceans, chitosan and its derivatives have been widely used for providing the safety of the foods (especially marine based foods) and shelf life extension. Thus, the study will be conducted.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

The proposed project focuses on the Utilization of Nanochitosan Powder for Bread Products (Pandesal and Ensaymada)

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Objectives: The general objective is to investigate the effect of nanochitosan powder on bread products. Specifically, it seeks to;

1. evaluate the sensory qualities of bread products with nanochitosan powder in terms of appearance, aroma, taste, texture and general acceptability.
2. determine the shelf-life of bread products with nanochitosan powder in terms of:
 - a. time and temperature
 - b. moisture content
 - c. pH level
 - d. molds and yeast
3. determine the product cost of sopa de fedio ala arroz caldo and ala chicken tinola.
4. find out the nutritional value of sopa de fedio ala arroz caldo and ala chicken tinola.

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(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

The present study will be based on the following prior arts such as:

Acceptable Amount of Chitosan in Food Applications

The acceptable amount of chitosan in most published articles usually added in food ranges from 0.5-2.0%. A study conducted by Park et al (2002b) on chitosan applied on baguette utilized 0.5%, 1.0% and 1.5% while Ahn et al. (2003) used 1% and 2% chitosan solution. In fruits and vegetables, an amount of 1% and 1.5% chitosan were used in the experimental study conducted by El Ghaouth et al. Darmadji and Izumimoto (1994a) also studied the effect of chitosan on the rancidity of meat with the application of 0.5% and 1%. A processed product, rice cake, was also treated with 1% and 2% chitosan in the study conducted by Leet et al. (2000).

Potential of Nanochitosan as Preservatives

Salt concentration combined with nano chitosan plays a very good role as an inhibitor in microbial activity during the drying process of fish, even with a relatively low concentration of nano chitosan of 0.4% (K E Savitri et al, 2021).

The shelf life of bread is generally limited due to staling and microbial growth (Leuschner et al., 1999). Staling is a general term that describes the time dependent loss in quality of flavor and texture of bread. Bread staling is a complex phenomenon in which multiple mechanisms operate (Gray & Bemiller, 2003).

Antimicrobial Activity of Chitosan

Chitosan has attracted attention as a potential food preservative of natural origin due to its antimicrobial activity against a wide range of foodborne filamentous fungi, yeast, and bacteria (Sagoo et al., 2002). There were several hypotheses on the mechanism of the microbial activity of chitosan. The most feasible hypothesis is a change in cell permeability due to interactions between the positively charged chitosan molecules and the negatively charged microbial cell membranes. This interaction leads to the leakage of proteinaceous and other intracellular constituents (Young et al., 1982; Leuba & Stossel, 1986; Papineau et al., 1991; Sudarshan et al., 1992; Fang et al., 1994). Other mechanisms are the interaction of diffused hydrolysis products with microbial DNA, which leads to the inhibition of the mRNA and protein synthesis (Hadwiger & Loschke 1981; Hadwiger et al., 1986; Sudarshan et al.,1992) and the chelation of metals, spore elements, and essential nutrients (Cuero et al., 1991).

(12) Scientific Basis/Theoretical Framework (Include concepts. theories and variables that indicate relevant relationships to be tested.

The product development include the utilization of nanochitosan powder for bread products, shelf-life testing, nutrient-content and microbial analysis and package development.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

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The followings parameters to be evaluated are:

- Sensory Evaluation
 - 10 experts and 32 panelists will evaluate the product in terms of its appearance, aroma, taste, texture, flavor, and general acceptability. A Nine(9)-point Hedonic scale will be used for sensory evaluation.
- Physico-chemical properties
 - pH, moisture content, water activity
- Shelf-life analysis
 - Molds and Yeast count in varying temperatures (room temperature)
- Nutritional Analysis

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	2024											
	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec
1. Gathering and reading materials connected to the study of interest												
2. Writing of Proposal												
3. Product Formulation and Development												
4. Product Evaluation												
5. Nutrient Content, Microbial Analysis and BFAD approved												
6. Analysis of Findings and writing of Data Gathered												
7. Colloquium /Defenses												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

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The expected output of this study will be the Development of Bread Products with Nanochitosan Powder.

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

The quality standards, cooking guide, and other IEC materials will be developed and produced. It will be disseminated and distributed to various interested stakeholders.

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The target beneficiaries of this study will be the following: bakery industry, bakers, consumers, and other food industries.

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Dr. Ace D. Barredo	Project leader	Food

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

Sources of Expenses	Quantity	Amount
A. Ingredients for pandesal and ensaymada.		Php 10,000.00
Nanochitosan powder		20,000.00
B. Materials, Tools, and Equipment		
LPG refill	2 tanks	2,000.00
Bond paper substance 20 8x11 ½	5 rims	930.00
Printer ink assorted colors	3 sets	2,370.00
Photo paper	10 sets	1,200.00
C. Nutrient-Content, Microbial Analysis, and BFAD Approved, Shelf-life test		40,000.00
D. Packaging		10,000.00

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E. English Editor's Fee		5,000.00
F. Publication		20,000.00
G. Presentation		20,000.00
H. Miscellaneous		5,000.00
Total Expenses		Php 136,500.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from **CAPSU Research Journal**, <http://researchjournal.capsu.edu.ph/>)

Sadik Büyükyörük. Chitosan for Using Food Protection, Submitted: September 14th, 2020 Reviewed: July 5th, 2021 Published: September 29th, 2021 <https://www.intechopen.com/chapters/78259>

Anand,M, P. Sathyapriya, M. Maruthupandy, A. Hameedha Beevi (2018).

Synthesis of chitosan nanoparticles by TPP and their potential mosquito larvicidal application, *Frontiers in Laboratory Medicine*, Volume 2, Issue 2, Pages 72-78, ISSN 2542-3649, <https://doi.org/10.1016/j.flm.2018.07.003>.

Acharya, V., A. Ghosh, AR Chowdhury,& Datta, P. (2022) Tannic acid-crosslinked chitosan matrices enhance osteogenic differentiation and modulate epigenetic status of cultured cells over glutaraldehyde crosslinking. *Soft Materials* 20:2, pages 149-160. *Artificial Cells, Nanomedicine, and Biotechnology*. An International Journal. Volume 46, 2018 - Issue sup3

Butler, B.L.; Vergano, P.J.; Testin, R.F.; Bunn, J.M.; et al. (1996). Mechanical and barrier properties of edible chitosan films as affected by composition and storage. *Journal of food science (USA)*. ISSN : 0022-1147. *Anim. Sci. Technol. (Jpn.)* 65 (7): 636-646, 1994

El-Naggar, N.EA., Saber, W.I.A., Zweil, A.M. *et al.* An innovative green synthesis approach of chitosan nanoparticles and their inhibitory activity against phytopathogenic *Botrytis cinerea* on strawberry leaves. *Sci Rep* 12, 3515 (2022). <https://doi.org/10.1038/s41598-022-07073-y>

Kim, Kyung-Hee, K, and Cho, Hee-Sook (2008). Physicochemical and Microbiological Properties of Skate (Raja kenoei) Kimchi on the Market. *Journal of the Korean Society of Food Culture (한국식생활문화학회지)*. Volume 23 Issue 2 Pages.235-242. 1225-7060(pISSN) 2288-7148(eISSN)

(20) Capsule Curriculum Vitae (One-page brief C.V. only)

ACE DISTOR BARREDO, was born on June 07, 1988 at Roxas Memorial Provincial Hospital, Roxas City, Capiz. He spent his elementary years at Plaridel East Elementary School, and his secondary education at Capiz National High School. He obtained his bachelor's degree at Capiz State University with the degree of Bachelor of Science in Industrial Education major in Foods as Cum-luade. He finished his master's and doctorate degree at the same university. Presently, he is an Assistant Professor I handling major and professional subjects under Industrial Technology Department.